



Silicon photonic dual-drive MIM based 56 Gb/s DAC-less and DSP-free PAM-4 transmission

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Abstract: We report two designs of silicon photonic dual-drive Michelson interferometric modulators (MIMs) suitable for four-level pulse amplitude modulation (PAM-4) that do not require digital-to-analog converters or digital signal processing. The PN junctions in MIM-1 have an asymmetric geometry and 4 doping concentrations, while those in MIM-2 have a symmetric geometry and 6 doping concentrations. We simulate and experimentally demonstrate that MIM-2 has a larger modulation efficiency and a better electro-optic (EO) bandwidth than MIM-1. The measured $V_{\pi}L_{\pi}$ of MIM-2 at -2 V bias is 0.8 V-cm, and the measured 3-dB EO bandwidth at 0 V bias is 9.3 GHz. By carefully choosing the bias conditions of the device and the driving binary radio-frequency signals applied on each phase shifter, PAM-4 signals with even spacings are generated. Successful 56 Gb/s PAM-4 transmission over 2 km of standard single mode fiber is presented, with an estimated bit error rate below the hard-decision forward error correction threshold of 3.8×10^{-3} .

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1. Introduction

Driven by the exponentially increasing demand of web media applications and cloud-based services, intra-data center traffic is growing rapidly [1]. To fulfill this challenge, high-speed short-reach optical transmission links with low power consumption have attracted immense research interest. Because intensity-modulation/direct-detection (IM/DD) systems do not require a local oscillator or complicated digital signal processing (DSP) [2], they are cost-effective for intra-data center applications. For 40G and 100G Ethernet applications, on-off keying (OOK) modulation has been selected [3]. However, a larger bandwidth is required for the next generation 400G Ethernet, therefore higher order intensity modulation formats are preferred. Among them, four-level pulse amplitude modulation (PAM-4) is a good choice when considering both spectral efficiency and signal-to-noise ratio [4]. Currently, $8\lambda \times 50\text{G}$ PAM-4 over 2 km of standard single mode fiber (SSMF) transmission (400GBASE-FR8) has been accepted for the next generation 400G Ethernet [5]. Silicon photonics is a mature CMOS-compatible platform, which enables a large volume integration with a low cost [6]. Therefore, it offers a promising solution for developing cost-effective data center interconnects. Recently, silicon photonic (SiP) modulators for PAM-4 signals transmission over distances suitable for intra-data center applications [7–9] have been reported.

For PAM-4 generation, electrical digital-to-analog converters (DACs) are typically used to drive the transmitters [10–12]. This method is direct and convenient, but largely increases the power consumption of the optical transmission links. Therefore, PAM-4 signal generation without using DACs or DSP in the entire link (e.g. pre-emphasis, pulse shaping, equalization, etc.) is attractive to reduce the power consumption. This has been achieved with various SiP modulators, such as segmented silicon-insulator-silicon capacitor (SISCAP) modulators [13], travelling-wave Mach-Zehnder modulators (TWMZMs) [14–17], cascaded dual microring modulators (MRMs) [18], multi-segmented inter-coupling ring modulators [19], ring-assisted Mach-Zehnder interferometers (RAMZIs) [20], two-segmented MRMs [21], multi-segmented MRMs [22], and dual parallel GeSi electro-absorption modulators (EAMs) [23]. The performances of these modulators is summarized in Table 1. The SISCAP modulators require a more complicated fabrication process than other modulators because of the vertical PN junctions, and their optical insertion loss (IL) is higher due to the poly silicon layer in the structure [13]. The TWMZMs occupy large footprints as their lengths are typically a few millimeters [14–17]. The MRMs are very sensitive to fabrication variations and temperature changes, so they need heaters to tune their resonant wavelengths [18–22]. The GeSi EAMs typically work in L-band (around 1600 nm) [23] because Ge has large absorption coefficients in O-band (around 1310 nm) and C-band (around 1550 nm) [24]. In addition, the dominant source of power consumption in TWMZMs is the termination resistor, and in the MRMs, it is the fast and accurate loop for thermally controlling resonances. A lumped SiP Michelson interferometric modulator (MIM), which avoids these additional sources of power consumption, was reported in [25]. Based on a single-drive configuration, 40 Gb/s OOK modulation was achieved [25].

In this paper, we present two designs of SiP dual-drive MIMs for DAC-less and DSP-free PAM-4 signal generation. An initial result of our work has been presented in [26]. The PN junctions in MIM-1 have an asymmetric geometry and 4 doping concentrations, while those in MIM-2 have a symmetric geometry and 6 doping concentrations. Both simulation and experiment results show that MIM-2 has a larger modulation efficiency and a better electro-optic (EO) bandwidth than MIM-1. The measured $V_{\pi}L_{\pi}$ of MIM-2 at -2 V bias is $0.8\text{ V}\cdot\text{cm}$, and the 3-dB EO bandwidth at 0 V bias is 9.3 GHz . Based on the measured direct-current (DC) transmission spectra, the bias conditions of the MIM and the binary radio-frequency (RF) signals applied on each phase shifter are carefully chosen to generate PAM-4 signals

with even spacings. Using MIM-2, PAM-4 transmission at 56 Gb/s over 2 km of SSMF is achieved, with an estimated bit error rate (BER) below the hard-decision (HD) forward error correction (FEC) threshold of 3.8×10^{-3} . Table 1 compares the previously published DAC-less and DSP-free PAM-4 SiP modulators [13–23] with this work, in which peak-to-peak voltages (V_{pp}), pseudo-random binary sequence (PRBS) lengths, bit rates, transmission distances and BERs are listed. It should be noted in [14,15,20] and this work, BERs are estimated using histograms of the captured eye diagrams based on a model where noises have Gaussian distribution [27,28]; while in [16–18,23], BERs are measured by direct error counting using real-time oscilloscopes (RTOs).

Table 1. Comparison of previously published DAC-less DSP-free PAM-4 SiP modulators with this work.

Reference	Modulator structure	V_{pp}	PRBS	Bit Rate	Transmission	BER
[13]	Segmented SISCAP modulators	2 V, 2 V	31	56 Gb/s	0	N. A. *
[14]	Segmented TWMZM	4.8 V, 4.8 V	31	100 Gb/s	1 km	$< 3.8 \times 10^{-3}$ **
[15]	Dual parallel TWMZMs	2 V, 4 V	31	100 Gb/s	2 km	$< 4.4 \times 10^{-3}$ **
[16]	TWMZM	1.5 V, 3 V	31	50 Gb/s	0	2.2×10^{-6} ***
[17]	TWMZM	5 V	15	64 Gb/s	0	6.6×10^{-5} ***
[18]	Cascaded dual MRMs	1 V, 1 V	9	30 Gb/s	0	4×10^{-3} ***
[19]	Multi-segmented inter-coupling ring modulator	3 V, 3 V	31	25 Gb/s	0	N. A. *
[20]	RAMZI	3.19 V, 4.26 V	31	50 Gb/s	2 km	3.3×10^{-3} **
[21]	Segmented MRM	4.4 V, 4.4 V	15	40 Gb/s	0	N. A. *
[22]	Multi-segmented MRM	1.55 V	31	40 Gb/s	0	N. A. *
[23]	Dual parallel GeSi EAMs	1.1 V, 2.2 V	7	112 Gb/s	2 km	1.4×10^{-3} ***
This work	Dual-drive MIM	1.41 V, 4.92 V	31	56 Gb/s	2 km	1.4×10^{-3} **

* BERs not reported.

** BERs estimated by using histograms of the captured eye diagrams.

*** BERs measured by error counting using RTOs.

2. PN junction design and simulation

Figure 1(a) shows an asymmetric PN junction design with 4 doping concentrations, which is the same as that in [25]. The 500-nm $\times$ 220-nm optical waveguide is laterally PN doped with a 100-nm offset, which means the width of the P doping region is 350 nm and that of the N doping region is 150 nm. The low doping concentrations in the rib waveguide are used to reduce the losses from optical scattering with the implanted carriers. The 6.35- μ m P++ and N++ doping areas in the 90-nm slab waveguide are for Ohmic contacts with aluminum pads. They are both 1 μ m away from the edge of the rib waveguide. In [25], the simulated peak doping concentrations are $7.8 \times 10^{17} \text{ cm}^{-3}$, $2.1 \times 10^{18} \text{ cm}^{-3}$, $3.9 \times 10^{19} \text{ cm}^{-3}$ and $9.7 \times 10^{19} \text{ cm}^{-3}$ for P, N, P++ and N++, respectively. Because the change in the concentration of holes contributes more to the plasma dispersion effect [29], the asymmetric PN junction in [25] was designed to increase the overlap of the optical mode with the P doping area in the rib waveguide. Similar asymmetric PN junction designs have also been reported in [30,31] because the doping concentration of the N region are both larger than that of the P region in these two references.

However, the doping concentrations used in the IME multi-project-wafer (MPW) run, which is the fabrication process of our designed devices, are different from those in [25,30,31]. As reported in [32], they are $5 \times 10^{17} \text{ cm}^{-3}$ for P, $3 \times 10^{17} \text{ cm}^{-3}$ for N, $2 \times 10^{18} \text{ cm}^{-3}$ for P+, $3 \times 10^{18} \text{ cm}^{-3}$ for N+, and $1 \times 10^{20} \text{ cm}^{-3}$ for both P++ and N++. As the P

doping concentration is already larger than the N doping concentration, the asymmetric design shown in Fig. 1(a) is not necessary. In this case, to improve the modulation efficiency, the overlap between the carrier depletion region and the optical mode should be optimized. Since the optical mode is confined in the middle of the rib waveguide, the carrier depletion region should also be centered in the middle of the rib waveguide. This can be achieved by designing the PN junction to be symmetric. When applying reverse bias voltages to deplete the carriers, the symmetric PN junction has a larger effective index change, and thus a larger modulation efficiency than the asymmetric one. As shown in Fig. 1(b), the rib waveguide is symmetrically PN doped, which means the widths of the P and N doping regions are both 250 nm. Furthermore, to decrease the junction resistance, two more doping concentrations of P + and N + are applied in the slab waveguide. In our design, the 810-nm P + and N + doping regions are 120 nm away from the rib waveguide edge. To keep the total width of the PN junction at 15.2 μm , which is the same as that shown in Fig. 1(a), the P ++ and N ++ doping regions are 6.42 μm .

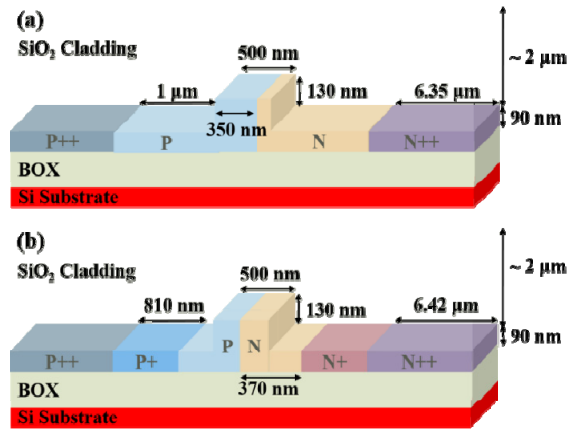


Fig. 1. (a) An asymmetric PN junction with 4 doping concentrations [25] and (b) a symmetric PN junction design with 6 doping concentrations.

We simulated these two different PN junction designs in Lumerical Device and MODE Solutions based on the constant doping concentrations in [32] and using the method in [33]. As shown in Figs. 1(a) and 1(b), there are two major differences between these two PN junction designs. One is the asymmetric PN doped region in Fig. 1(a) is redesigned to be symmetric in Fig. 1(b). The other difference is the number of doping concentrations, which are 4 in Fig. 1(a) and 6 in Fig. 1(b). Therefore, to investigate the specific reasons of the differences in their simulated performance, we also simulated a PN junction with a symmetric geometry and 4 doping concentrations shown in Fig. 2.

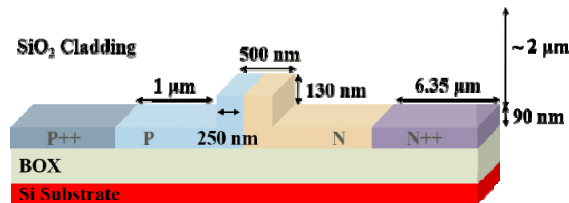


Fig. 2. Symmetric PN junction design with 4 doping concentrations.

Figure 3(a) shows the simulated effective index changes at various reverse bias voltages. As analyzed, they are larger in the symmetric PN junctions than those in the asymmetric one because of more overlapping of the optical mode with the carrier depletion region. In

addition, adding P + and N + doping areas in the slab waveguide does not further improve the effective index changes, as shown by the blue and green curves in Fig. 3(a). The simulated propagation losses are shown in Fig. 3(b). Since the P doping concentration is much larger than the N doping concentration, reducing the width of P area by 100 nm results in less optical scattering in the rib waveguide. Therefore, the simulated propagation losses are smaller in the PN junction with symmetric 4 doping concentrations than those in the PN junction with asymmetric 4 doping concentrations. But adding P + and N + regions in the slab waveguide increases the propagation losses, as there are more carriers and the optical scattering loss is larger. From the comparison of the red and green curves in Fig. 3(b), we can see that when the applied reverse bias voltage is smaller than 4 V, the simulated propagation losses are larger in the PN junction with symmetric 6 doping concentrations due to more implanted carriers; however, when the applied reverse bias voltage is larger than 4 V, the losses are smaller than those in the PN junction with asymmetric 4 doping concentrations due to more effective carrier depletion.

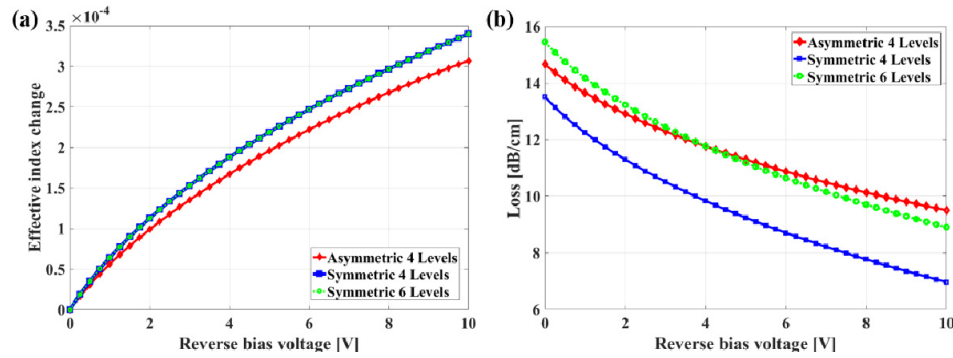


Fig. 3. Simulated (a) effective index changes and (b) propagation losses of the PN junctions with asymmetric 4, symmetric 4 and symmetric 6 doping concentrations.

The P + and N + doping regions are added in the slab waveguide to decrease the junction resistance. As listed in Table 2, the simulated junction resistances do not change much whether the PN junction is asymmetric or symmetric, but they are reduced by more than 50% with P + and N + implantations in the slab waveguide. This is beneficial for increasing the 3-dB cutoff frequency of the phase shifter, which results in a better EO bandwidth of the device.

Table 2. Simulated resistances ($\Omega\text{-cm}$) of the three PN junctions.

	-2 V	-4 V	-6 V	-8 V	-10 V
Asymmetric 4 doping concentrations	1.3542	1.3535	1.3525	1.3510	1.3492
Symmetric 4 doping concentrations	1.3373	1.3364	1.3351	1.3335	1.3316
Symmetric 6 doping concentrations	0.6258	0.6236	0.6211	0.6184	0.6158

By comparing the simulated results of the PN junctions with symmetric 4 and 6 doping concentrations shown in Fig. 3(b) and Table 2, introducing the P+/N + doping regions in the slab waveguide results in larger propagation losses and smaller PN junction resistances. To make a trade-off between them, numerical simulations are needed to optimize the width of the P+/N + doping regions. Based on the constant doping concentrations in [32] and the same method in [33], the simulated results are shown in Fig. 4. Starting from a P+/N + region width of approximately 800 nm, the propagation loss increases significantly, as shown in Fig. 4(a). The PN junction resistance decreases almost linearly with the P+/N + region width, as shown in Fig. 4(b). In our design, we used a P+/N + doping region width of 810 nm in the slab

waveguide so that the propagation losses are no more than 2 dB/cm larger than those without applying P+/N+ doping, meanwhile the PN junction resistance decreases more than 50%.

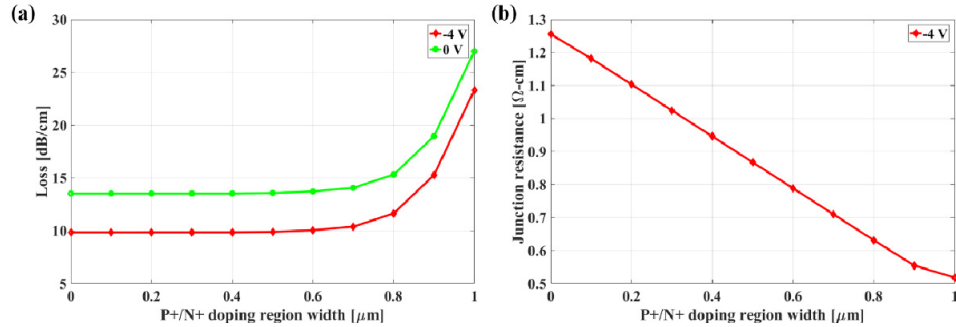


Fig. 4. Simulated (a) propagation losses at 0 V and -4 V bias voltages, (b) PN junction resistances at -4 V bias voltage with different widths of the P+/N+ doping regions.

In conclusion, based on the doping concentrations used in the fabrication processes [32], we designed a symmetric PN junction with 6 doping concentrations. Compared to the symmetric PN junction design with 4 doping concentrations [25], our design has a larger simulated modulation efficiency and smaller simulated junction resistances, which are both favored for a better modulation performance.

3. Device design and fabrication

The schematic layout of the SiP dual-drive MIM is shown in Fig. 5. After coupling into the device using a grating coupler (GC), light travels through a 2×2 3-dB adiabatic coupler [34] and followed by two arms with an imbalance of approximately 200 μm . The 500- μm phase shifters are designed for modulation in the carrier-depletion mode. To achieve dual-drive operation, both phase shifters have metallization for applying independent binary RF signals. Because of the loop mirrors at the end of the modulator, light is coupled back and passes through the two arms twice. Therefore, the modulation efficiency of the MIM is doubled. After travelling through the adiabatic coupler again, light is coupled out of the device by a GC. Based on the two PN junction designs shown in Fig. 1 and the same structure shown in Fig. 5, two MIMs are designed. The PN junctions in MIM-1 have an asymmetric geometry and 4 doping concentrations, while those in MIM-2 have a symmetric geometry and 6 doping concentrations.

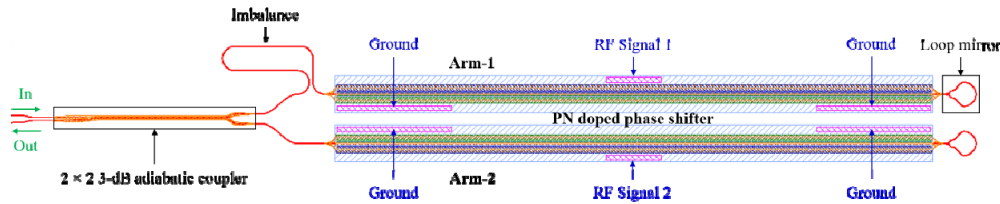


Fig. 5. Schematic layout of the SiP dual-drive MIM.

For a fair comparison, the two MIMs were fabricated on one chip in the same MPW run at IME. The micrographs of the fabricated devices are shown in Figs. 6(a) and 6(b), respectively. For a dual-drive operation, each phase shifter has a metal pad for applying independent RF signals, and all the ground pads are connected. The total footprint of each device, including the metal bond pads, is $830 \mu\text{m} \times 430 \mu\text{m}$.

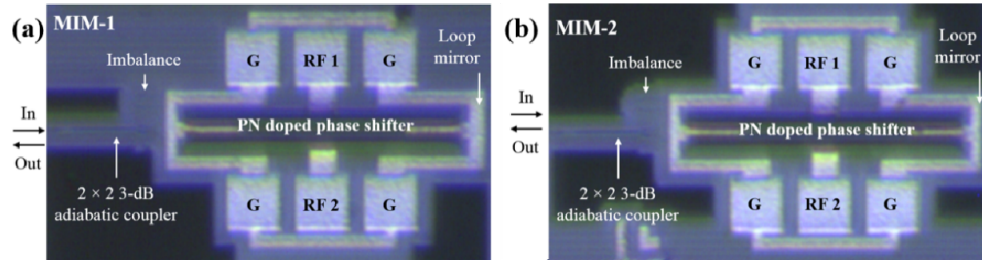


Fig. 6. Micrographs of the fabricated (a) MIM-1 and (b) MIM-2.

4. Device characterization

4.1 DC characterization

Figure 7 shows the measured DC transmission spectra when applying reverse bias voltages on each arm of MIM-1. At 0 V, the total on-chip IL is 13.73 dB. Excluding the 9.85-dB IL from the GC pair and the 1.30-dB IL from the routing waveguides, the IL of MIM-1 is 2.58 dB. The measured free spectral range (FSR) is 1.43 nm. This corresponds to an effective waveguide imbalance of 400 μm . Since light makes two passes through the phase shifters and the two arms have an imbalance of 200- μm , the measured FSR is in good agreement with the device design.

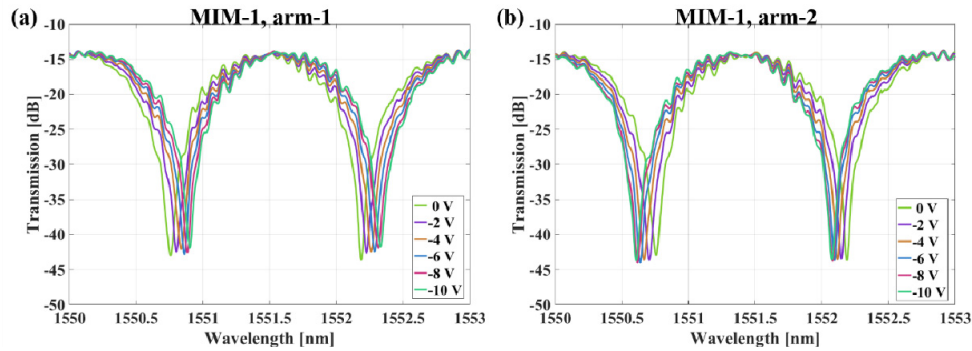


Fig. 7. Measured DC transmission spectra when applying reverse bias voltages on (a) arm-1 and (b) arm-2 of MIM-1.

The measured DC transmission spectra of MIM-2 are shown in Fig. 8. The total on-chip IL is 13.78 dB, and the routing waveguide has an IL of 1.10 dB. MIM-2 has a measured IL of 2.83 dB, which is larger than MIM-1 due to the larger propagation losses in the waveguide, as shown in Fig. 3(b). The measured FSR is 1.44 nm, and it also corresponds to an effective waveguide imbalance of 400 μm . In addition, the wavelength shifts of MIM-2 are much larger than those of MIM-1, because the modulation efficiencies of the symmetric PN junctions are larger than those of the asymmetric ones, as shown in Fig. 3(a).

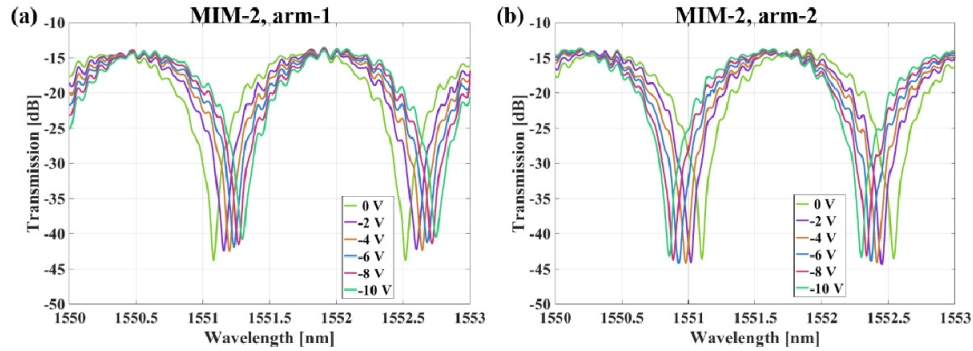


Fig. 8. Measured DC transmission spectra when applying reverse bias voltages on (a) arm-1 and (b) arm-2 of MIM-2.

The $V_{\pi}L_{\pi}$ of each phase shifter is calculated based on the measured DC transmission spectra. Under a reverse bias voltage of V_{bias} , the phase shift is $\Delta\phi = 2\pi\Delta\lambda/\text{FSR}$, where $\Delta\lambda$ is the wavelength shift at V_{bias} . The calculated phase shift of each arm on the two MIMs are shown in Fig. 9(a). $V_{\pi}L_{\pi} = \pi V_{\text{bias}}L/\Delta\phi$ [35], where $L = 500 \mu\text{m}$ is the phase shifter length. As shown in Fig. 9(b), at a bias voltage of -2 V , $V_{\pi}L_{\pi \text{ MIM-1, arm-1}} = 1.6 \text{ V-cm}$, $V_{\pi}L_{\pi \text{ MIM-1, arm-2}} = 1.4 \text{ V-cm}$, $V_{\pi}L_{\pi \text{ MIM-2, arm-1}} = 0.9 \text{ V-cm}$, and $V_{\pi}L_{\pi \text{ MIM-2, arm-2}} = 0.8 \text{ V-cm}$. The MIM-2 has a smaller $V_{\pi}L_{\pi}$ than MIM-1 because of the larger modulation efficiency, which was shown in the simulation of Fig. 3(a) and confirmed by the measured DC transmission spectra in Figs. 7 and 8. When applying higher reverse bias voltages, the $V_{\pi}L_{\pi}$ becomes larger due to the diminishing phase shifts. It should be noted that although the phase shifters on each MIM are designed to be identical, their measured $V_{\pi}L_{\pi}$ are different mainly due to the variations in the fabrication process. The measured $V_{\pi}L_{\pi}$ of the two MIMs on the other die also show deviations between the two arms, which confirms the consistency of the results in the presence of fabrication variations. Similar fabrication variations have also been reported in [8, 9, 14]. In [36], the characterization of this manufacturing variability and the prediction of the correlated circuit performance have been addressed.

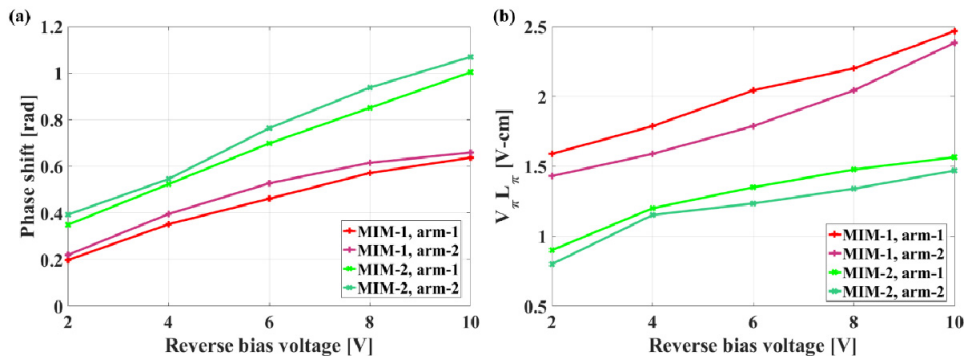


Fig. 9. Calculated (a) phase shifts and (b) $V_{\pi}L_{\pi}$ of each phase shifter on the two MIMs.

For DAC-less and DSP-free PAM-4 signal generation, independent binary RF signals are applied on each phase shifter of the MIMs. To achieve even spacings between the adjacent levels of the generated PAM-4 signals, the bias conditions and the parameters of the applied RF signals should be carefully chosen. Figure 10 shows the measured output power when applying reverse bias voltages on the RF pads of both phase shifters simultaneously. The input power from an external cavity laser (ECL) is 15.5 dBm. For a fair comparison, the voltages applied on the two MIMs are kept the same. In addition, the bias wavelengths are chosen to assure that the two MIMs have comparable output powers at their ‘11’ levels. As

shown in the inset of Fig. 10(a), the four output powers of MIM-1 at 1550.67 nm are 0.328 mW, 0.303 mW, 0.280 mW and 0.257 mW. For MIM-2, a bias wavelength of 1550.9 nm is chosen so that the output powers are 0.321 mW, 0.258 mW, 0.203 mW and 0.146 mW, as shown in the inset of Fig. 10(b). The four power levels of the two MIMs both have reasonably even spacings. The average spacing of MIM-2 is 0.058 mW, which is larger than the 0.024 mW of MIM-1. Based on the voltages shown in Fig. 10, which are $V_{1,1} = -3.53$ V, $V_{1,2} = -2.12$ V, $V_{2,1} = -2.93$ V, and $V_{2,2} = -7.85$ V, the bias voltages and the peak-to-peak voltages of the binary RF signals applied on each phase shifter are chosen accordingly, which are $V_{\text{bias},1} = -2.83$ V, $V_{\text{pp},1} = 1.41$ V, and $V_{\text{bias},2} = -5.39$ V, $V_{\text{pp},2} = 4.92$ V.

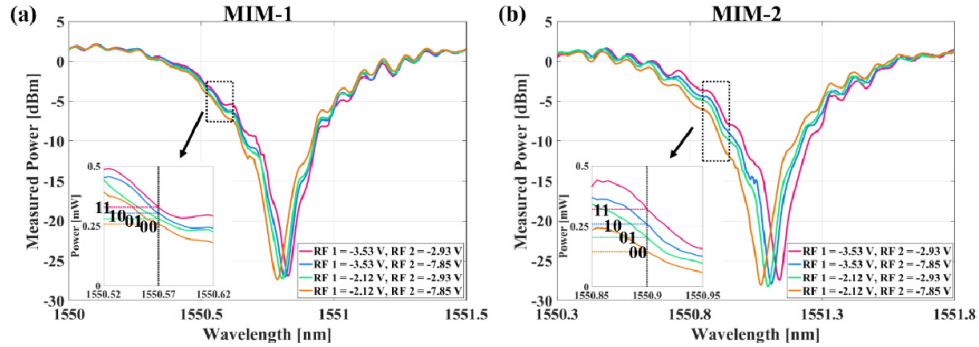


Fig. 10. Measured DC transmission spectra when applying reverse bias voltages on both phase shifters of (a) MIM-1 and (b) MIM-2.

As shown in Figs. 7, 8 and 10, there are ripples in the measured DC transmission spectra of both MIMs. This is because of the Fabry-Perot resonator formed by the output grating coupler which reflects a small portion of light back into the device, and the 2×2 3-dB adiabatic coupler in the MIM structure. As the two MIMs are biased at the wavelengths to avoid these ripples and to obtain four evenly spaced output powers, their dynamic ILs are high. Excluding the IL of the device itself, both MIMs have dynamic ILs of approximately 6.7 dB. By driving the two arms using RF signals with higher peak-to-peak voltages can decrease the dynamic IL of the MIMs, but in this case a larger power consumption is needed as a trade-off.

4.2 Small-signal characterization

The EE S_{11} and EO S_{21} responses of each phase shifter on the two MIMs were measured using an Agilent lightwave component analyzer after calibrating the cables and probes. Figure 11 shows the measured results under 0 V bias voltage at the modulation wavelengths shown in Fig. 10, i.e. 1550.57 nm for MIM-1 and 1550.9 nm for MIM-2. All the measured magnitudes are normalized at 1.5 GHz. The 3-dB EE and EO bandwidths of the two MIMs are listed in Table 3. As the only difference in the two designs is the PN junction design, the larger EE and EO bandwidths of the phase shifters on MIM-2 are due to the smaller junction resistances, as analyzed in section 2. However, though the PN junction resistances are reduced by more than 50%, the 3-dB EE bandwidths of MIM-2 are not twice of those of MIM-1. This is because in addition to the PN junction capacitance and resistance, there are other elements in the small signal model of an MIM [25], such as the capacitance of the contact pads [25] and the parasitic lead inductance [37]. In addition, there are differences between the measured responses of the phase shifters on each MIM. For example, at frequencies above 20 GHz, more attenuation and fluctuation are observed in the EE responses of arm-2 on MIM-2, as shown in Fig. 11(a). Therefore, its EO responses has larger attenuation at high frequencies, as shown in Fig. 11(b). Similar variations in the two phase

shifters on the same modulator have also been reported in [14], which is also due to the deviations in the fabrication process.

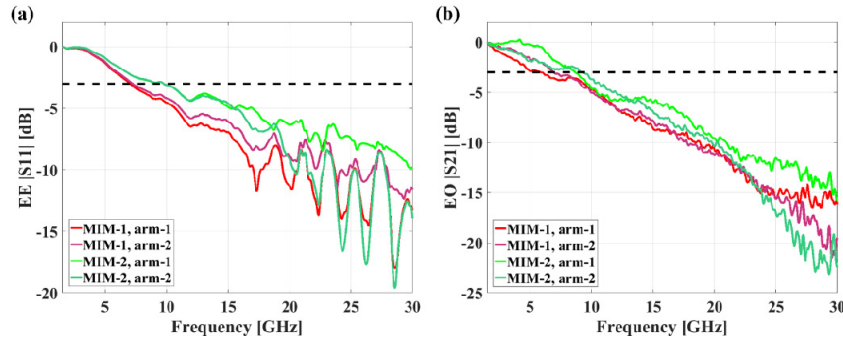


Fig. 11. Measured (a) EE S_{11} magnitudes and (b) EO S_{21} magnitudes under 0 V bias voltage.

To evaluate the performance of the devices at the operating bias points, we further investigated their EO responses at -2.83 V bias for arm-1 and -5.39 V bias for arm-2. As shown in Fig. 12 and listed in Table 3, the EO bandwidths are all larger than those at 0 V due to the reduction of junction capacitances at higher reverse bias voltages.

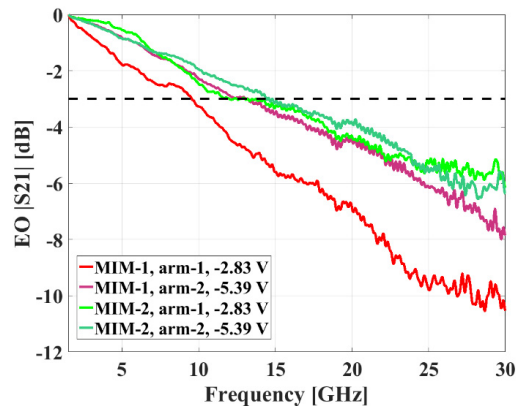


Fig. 12. Measured EO S_{21} magnitudes under -2.83 V for arm-1 and -5.39 V for arm-2.

Table 3. Measured 3-dB EE and EO bandwidths.

Device	Phase shifter	EE bandwidth (GHz)	EO bandwidth (GHz)
MIM-1	arm-1	7.0	5.9 (at 0 V)
			9.5 (at -2.83 V)
	arm-2	7.2	6.9 (at 0 V)
			13.1 (at -5.39 V)
MIM-2	arm-1	9.8	8.7 (at 0 V)
			12.0 (at -2.83 V)
	arm-2	9.8	9.3 (at 0 V)
			14.4 (at -5.39 V)

5. PAM-4 experiment and results

The experimental setup for DAC-less and DSP-free PAM-4 using the SiP dual-drive MIMs is shown in Fig. 13. Two PRBS-31 binary RF signals were generated from two independent

channels of a bit pattern generator (BPG), and were then amplified to the required peak-to-peak voltages ($V_{pp,1} = 1.41$ V and $V_{pp,2} = 4.92$ V) by 25-GHz RF amplifiers. The required bias voltages ($V_{bias,1} = -2.83$ V and $V_{bias,2} = -5.39$ V) together with the RF signals were applied on the phase shifters of the MIMs using 65-GHz bias tees and 40-GHz GSG probes. The input power of the ECL was set to be 15.5 dBm, and the wavelengths were 1550.57 nm for MIM-1 and 1550.9 nm for MIM-2. After modulation, the PAM-4 signals were transmitted in both the back-to-back (B2B) configuration and over 2 km of SSMF. Then the average optical power was amplified to 3 dBm by an erbium-doped fiber amplifier (EDFA). Afterwards, the eye diagrams were obtained by a 65-GHz digital communication analyzer (DCA) optical module.

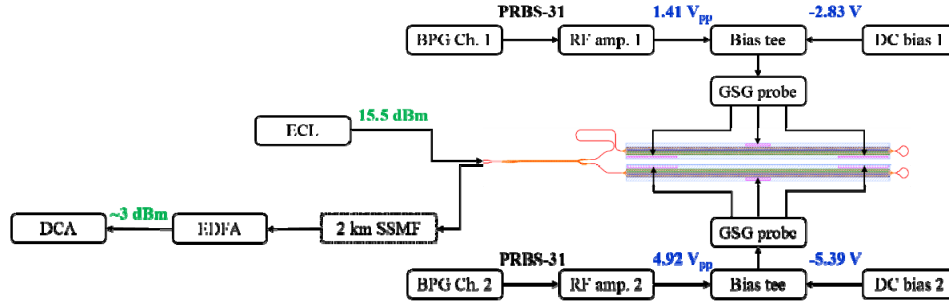


Fig. 13. Experimental setup for DAC-less and DSP-free PAM-4 using the SiP dual-drive MIMs.

Based on the assumption that the noise has a Gaussian distribution, the PAM-4 BERs are estimated using the vertical histograms of the measured eye diagrams at every bit rate and every transmission distance [27, 28]. In [28], this BER estimation method is explained in detail, and good agreement has been observed between the estimated BERs and those obtained by direct error counting. As shown in Fig. 14, the estimated BERs of MIM-1 from 40 Gb/s to 60 Gb/s are all higher than the KP4 FEC threshold of 2.2×10^{-4} , and the BER at 50 Gb/s in the B2B configuration is below the HD FEC threshold of 3.8×10^{-3} .

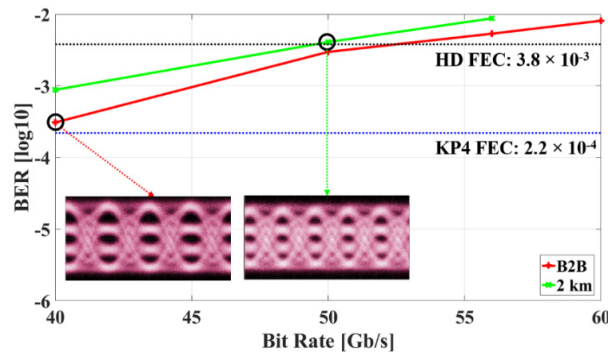


Fig. 14. PAM-4 BERs and eye diagrams of MIM-1.

Figure 15 shows the estimated BERs of MIM-2. The measured eye diagrams of MIM-2 at 40 Gb/s in the B2B configuration and at 50 Gb/s after 2 km of SSMF transmission are both clearly open. They are much better than those of MIM-1 shown in Fig. 14. After transmission over 2 km of SSMF, the BER of MIM-2 at 50 Gb/s is below the KP4 FEC threshold, and it is below the HD FEC threshold at 56 Gb/s. By turning on the RF signals on each phase shifter one at a time, the 28-Gbaud eye diagrams after transmission over 2 km of SSMF are shown in Figs. 13(b) and 13(c), respectively. Their measured optical modulation amplitudes (OMAs) are 0.68 mW in arm-1 and 1.20 mW in arm-2. The PAM-4 optical signals are generated by

combining the two binary signals modulated by each phase shifter. The measured four modulated power levels of the 56 Gb/s PAM-4 eye diagram after 2 km of SSMF transmission are 3.01 mW, 2.35 mW, 1.73 mW and 1.19 mW. They have reasonable even spacings, as shown in Fig. 15(a). The better PAM-4 results of MIM-2 are predominantly because of the larger modulation efficiencies and better EO bandwidths, as analyzed in section 2 and measured in section 4.

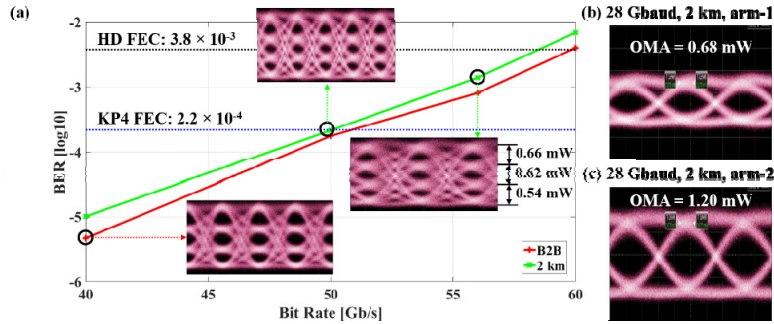


Fig. 15. (a) PAM-4 BERs and eye diagrams of MIM-2, 28-Gbaud eye diagrams after 2 km of SSMF transmission by tuning on the RF signals on (a) arm-1 and (b) arm-2 of MIM-2.

6. Conclusion

We present two designs of SiP dual-drive MIMs for DAC-less and DSP-free PAM-4 signal generation. Based on the doping concentrations applied in the fabrication process, the PN junctions with symmetric geometry and 6 doping concentrations have larger simulated modulation efficiencies and smaller simulated junction resistances than those with asymmetric geometry and 4 doping concentrations. MIM-2, designed with PN junctions of symmetric 6 doping concentrations, has a measured $V_{\pi}L_{\pi}$ of 0.8 V-cm at -2 V bias, and a measured 3-dB EO bandwidth of 9.3 GHz at 0 V bias. They are better than those of MIM-1, which is designed with PN junctions of asymmetric 4 doping concentrations. When applying bias voltages on both phase shifters, the measured DC transmission spectra show four output power levels at the modulation wavelengths. The bias conditions and driving RF signals are chosen accordingly to achieve PAM-4 with reasonable even spacings. Driven by the same RF signals and biased at similar conditions, the PAM-4 transmission results of MIM-2 are better than those of MIM-1. After 2 km of SSMF transmission, MIM-2 has an estimated BER at 50 Gb/s below the KP4 FEC threshold, and at 56 Gb/s it is below the HD FEC threshold. The operating principle and the experimental results shown in this paper show that the SiP dual-drive MIM is a good candidate for the next generation cost-effective 400G Ethernet.

The MIM structure is basically a folded MZM structure. Comparing to a MZM with 1-mm phase shifter and lumped electrode design [38], the main limiting factor of the MIM is the high dynamic IL resulted from biasing it at a wavelength to avoid the ripples in the transmission spectra, as analyzed in section 4.1. The EO bandwidth of the MIM, predominantly limited by the phase shifter length, is relatively small, especially when comparing to the MZMs with travelling-wave electrodes and phase shifters of several millimeters [8, 9, 14–17]. Therefore, to further improve the modulation performance of the MIM, a trade-off between the EO bandwidth and the phase shifter length should be considered.

Acknowledgments

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